



Bertran Miquel Oliver

PhD Student in AI

- November 3rd, 1996
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Links

Personal webpage
LinkedIn profile

Languages

- | | | |
|--|---------|-----------|
| | Catalan | ● ● ● ● ● |
| | English | ● ● ● ● ● |
| | French | ● ● ● ● ● |
| | German | ● ● ● ● ● |
| | Spanish | ● ● ● ● ● |

Technical skills

- | | | |
|--|-------------|-----------|
| | Python | ● ● ● ● ● |
| | Git | ● ● ● ● ● |
| | LaTeX | ● ● ● ● ● |
| | R | ● ● ● ● ● |
| | SQL | ● ● ● ● ● |
| | C++ | ● ● ● ● ● |
| | C | ● ● ● ● ● |
| | Mathematica | ● ● ● ● ● |

ML frameworks: PyTorch, PyTorch Geometric
HPC/tools: Linux, SLURM, Weights & Biases

Work experience

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|-----------------------|--|
| Feb. 2024 - now | PhD Student
<i>BSC - Barcelona Supercomputing Center</i> |
| Sep. 2023 - Aug. 2025 | Part-time Lecturer
<i>UB - University of Barcelona</i>
Teaching undergraduate Computer Science courses. |
| June 2022 - June 2023 | Working student Data Analyst
<i>IFCO</i>
Providing data analysis to worldwide customers.
Participating on a simulation project with Python. |
| June 2020 - June 2021 | BI Technician
<i>Teladoc Health International</i>
Integrating different systems into a BI tool, managing with databases and creating data models. |
| Nov. 2019 - June 2020 | Business Intelligence Consultant
<i>SDG</i>
Supporting international pharmaceutical businesses to become data-driven companies, using Qlik and Tableau. |

Academic formation

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|--------------------------|---|
| April 2021 - August 2023 | MSc. Degree in Mathematics
<i>Minor in Data Science & Biomathematics</i>
TUM - Technical University of Munich
Final degree thesis: <i>Mathematical Investigation into Genomic Order of Different Vertebrate Species</i> - Grade: 1.0
Relevant courses: <i>Introduction to Deep Learning, Machine Learning, Mathematical Models in Biology, Genetics</i> |
| Sep. 2014 - June 2019 | BSc. Degree in Mathematics
UB - University of Barcelona |

Publications

- | | |
|------|---|
| 2026 | “Effective Resistance Rewiring: A Simple Topological Correction for Over-Squashing”
Miquel-Oliver et al., PMLR, ICLR 2026 GRaM Workshop. arXiv:2603.11944. |
| 2025 | “Character-level Tokenizations as Powerful Inductive Biases for RNA Foundational Models”
Morales-Pastor et al., LMRL and AI4NA Workshops, ICLR 2025, Singapore. arXiv:2411.11808. |
| 2025 | “TopoBench: A Framework for Benchmarking Topological Deep Learning”
Telyatnikov et al., DMLR, ICLR Vienna. arXiv:2406.06642. |